AI Co-Pilot: Telemetry Dataset & Machine Learning System Architecture

End-to-End Pipeline: Multi-Modal Ingestion, BiLSTM Action Classification, Continuous Risk Regression & Dual Anomaly Detection

TIER 1: MULTI-SOURCE AVIATION DATA INGESTION & FEATURE ENGINEERING PIPELINE

NASA ASRS Database
- 6,978 Cleaned Safety Incident Records
- Multi-system operational failure reports
- Flight crew human-factor context & narratives
- Target Labels: 9 Emergency SOP Classes

OpenSky Squawk 7700 Trajectories
- 832 Declared Emergency Flight Segments
- ADS-B transponder telemetry (1 Hz sampled)
- Dynamic 30-min temporal gap segmentation
- Kinematics: Altitude, groundspeed, track, Vz

Synthetic Normal Envelopes
- 1,716 Nominal Baseline Flight Profiles
- Bounded climb, cruise, and descent envelopes
- Clean negative-class baseline flight sequences
- Zero-alarm baseline for anomaly detectors

12-Parameter Aerodynamic Feature Synthesizer
- Kinematics: Altitude (h), Airspeed (Va), V-Rate (h_dot), Pitch (θ), Roll (ϕ), Track (ψ)
- Control & Energy: Engine (N1), Load (G), Dist (d_rw), Wind (Vw), Energy (E), Phase (κ)
- Zero-Leakage: Z-score (μ_train, σ_train) parameters fitted strictly on training partition

Sliding Window Tensor & Flight-Segment Partition
- Sequence Structuring: T = 15 Timesteps, Stride S = 5 → Tensor X ∈ ℝ^(B × 15 × 12)
- Segment-Level Split (80:20): 7,620 Train Sequences | 1,906 Held-Out Test Sequences
- Strict flight-segment isolation completely eliminates temporal autocorrelation leakage

TIER 2: MACHINE LEARNING & DEEP LEARNING MODEL ARCHITECTURES

MODEL 1: BiLSTM ACTION CLASSIFIER
Supervised Sequence Classification (9 SOP Classes)

Input Projection & Latent Space
- Input Tensor: X ∈ ℝ^(15 × 12)
- Projection: W_in ∈ ℝ^(12 × 64), b_in ∈ ℝ^64
- Latent: H^(0) = ReLU(X · W_in + b_in) ∈ ℝ^(15 × 64)
- Projects 12 aerodynamic channels to 64D space

Bi-Directional Temporal Mixing
- Forward Pass: h_fwd_t = ReLU(H_t + W_f · h_fwd_t-1)
- Backward Pass: h_bwd_t = ReLU(H_t + W_b · h_bwd_t+1)
- Fused Seq: H_seq = ½(H_fwd + H_bwd) ∈ ℝ^(15 × 64)
- Captures causal onset and forward trend dynamics

Recency-Biased Self-Attention
- Attention: a_t = v_att^{T} tanh(W_att · H_t) + λ_recency
- Softmax Weights: α = Softmax(a) ∈ ℝ^15
- Context Vector: c = ∑ α_t H_t ∈ ℝ^64
- Fused State: [c || h_15] ∈ ℝ^128 (Context + Current)

Classification Head & Performance
- Dense Layer: FC1 = ReLU([c || h_15] · W_fc1 + b_fc1)
- Softmax Head: P(y | X) = Softmax(FC1 · W_cls + b_cls)
- 9 SOP Directives: Stall, CFIT, Windshear, Flameout...
- Accuracy: 98.85% | AUC-ROC: 0.988 | F1: 98.81%
- Precision: 98.82% | Recall: 98.85% | Loss: 0.052

MODEL 2: CONTINUOUS RISK REGRESSOR
Physics-Guided Envelope & Energy Dynamics

Kinematic Boundary Penalties
- Sink Rate: r_vrate = min(40, (|h_dot|/4000)^1.8 · 40)
- Airspeed Decay: Vs < 115 kts, Vmo > 340 kts
- Pitch Limits: θ > 18° climb or θ < -10° dive
- Bank Excursion: Roll |ϕ| > 35° excessive bank

Aerodynamic Energy & Load Penalty
- Structural Load Factor: r_g = |G - 1.0| > 0.4
- Terrain Margin: Alt < 1,200 ft & Dist > 4 nm
- Severe Windshear: Wind vector > 35 kts
- Mech Energy Rate: dE/dt = h_dot + (Va/g)·Va_dot

Composite Hazard Formulation
- Aerospace Penalty: R_comp = ∑ (w_i · r_i), i ∈ {1..12}
- Severity Multiplier: Max(R_comp · 1.5, 75.0)
- Clamped Scale: R = min(100.0, max(5.0, R_comp))
- Standardized against FAA hazard envelopes

Risk Regression Performance
- Continuous Safety Index: R ∈ [0, 100] Points
- Goodness of Fit: R^2 = 0.972 (97.2% Variance)
- Errors: RMSE = 3.12, MAE = 2.50, MSE = 9.73
- Residual Errors: Zero-mean Gaussian (μ=0.04)
- Tiers: Low (<25), Med (25-50), High (50-75), Crit (≥75)

MODEL 3: DUAL ANOMALY DETECTORS
Unsupervised Trajectory Outlier & Discrepancy

Path A: LSTM Autoencoder (Recon)
- Encoder: LSTM(12 → 64) → Latent Code z ∈ ℝ^64
- Decoder: Repeat(15) → LSTM(64) → Linear(12)
- Reconstructed Output: X̂ ∈ ℝ^(15 × 12)
- Pointwise Error: MAE(X, X̂) = (1/TF) ∑∑ |x - x̂|

Path A Threshold & Validation
- Calibration Cutoff: τ_AE = 0.0825 (99th %ile)
- Held-Out Anomaly Rate: 1.41% (1,928 windows)
- Test Errors: MAE = 0.0163, RMSE = 0.0248
- Identifies sharp kinematic discontinuities

Path B: Anomaly Transformer (Assoc)
- Series Association: S_assoc ∈ ℝ^(15 × 15)
- Prior Association: P_assoc ∈ ℝ^(15 × 15)
- Association Discrepancy: D_KL(S || P) + D_KL(P || S)
- Detects breakdown in multi-sensor correlations

Path B Verification & Dual Fusion
- Anomaly Score: MSE · Softmax(-D_KL(S || P))
- Calibration Cutoff: τ_Trans = 0.003329 (99th %ile)
- Flagged Rate: 11.33% windows (15,552 flagged)
- Attribution: Isolates failing sensor telemetry
- Fusion: Pointwise MAE + Relational Shifts

TIER 3: IN-MEMORY NUMPY INFERENCE RUNTIME & REAL-TIME AVIONICS INTEGRATION

Pure NumPy In-Memory Inference Engine (Zero External Framework Overhead in Flight Loop)
- Cockpit Execution: Vectorized C-contiguous NumPy runtime with pre-allocated activation memory buffers
- Single-Window Latency: 2.1 ms | Peak P95 Latency: 2.6 ms | Throughput: 445 Windows/s (445 FPS)
- Avionics Budget: 20 Hz telemetry budget = 50 ms → Engine consumes only 4.2% of compute budget
- Resource Footprint: < 1.2 GB RAM | Deterministic IEEE-754 float32 matrix operations on edge avionics

1. SOP Recovery Directive
- Discrete Action Directive: P(y_k | X) ≥ 0.95
- 9 FAA Standard Operating Procedures
- Stall Recovery, CFIT Terrain Pull-Up, Windshear Escape, Flameout Restart, Overspeed Decel, Severe Turbulence
- Real-Time Pilot Advisory in < 3 ms

2. Continuous Risk Gauge
- Real-Time Risk Score: R ∈ [0, 100]
- Regression Fit: R^2 = 0.972, RMSE = 3.12, MAE = 2.50
- Four Operational Threat Severity Tiers: Nominal (<25), Moderate (25-50), Elevated (50-75), Critical Hazard (≥75)
- Zero-lag aerodynamic margin indicator

3. Dual Anomaly Verification
- Statistical Dual Alarm: Pointwise + Relational
- Dual Cutoffs: MAE (1.41%) + Assoc (11.33%)
- Calibration: Zero alarms on 1,716 normals
- Multi-Variate Telemetry Attribution: Pinpoints deviating sensor channels for immediate cockpit verification & logging